

Parabolas

Montague Hamilton

2 October 2026

1 Questions

1. **Describe fully and test:** $y = -3(x + 2)^2 + 5$ - **vertex, opening direction, shape and the y-intercept.**

The vertex will occur at $(-2, 5)$, as

$$y = -3((-2) + 2)^2 + 5$$

$$y = -3(0)^2 + 5$$

$$y = 5$$

The opening direction will be downwards, as the y values get inverted by the negative in front of the expression, $-3(x + 2)^2$. The shape will be quite narrow, as for every addition to x, y changes by 3 (shrinks in this case). The y-intercept occurs when $x = 0$, so

$$y = -3((0) + 2)^2 + 5$$

$$y = -3 \cdot 4 + 5$$

$$y = -7$$

The y-intercept occurs at $(0, -7)$.

2. **Describe fully:** $y = 2 \sin(3x) - 4$ - **amplitude, midline and period.**

The amplitude will be 2, as for every value of $\sin(3x)$, y is twice as big as otherwise. The midline will be -4 , as the last number added or subtracted from the rest of the function shifts every y output by that amount. The frequency is 3, so the period is $\frac{2\pi}{3}$

3. **Spot the error: a student says $y = (x + 3)^2$ has vertex (3,0). Test the claim, state the true vertex, and name the rule they violated.**

At $x = 3$,

$$y = ((3) + 3)^2$$

$$y = 36$$

At $x = 2$, $y = 25$ and at $x = 4$, $y = 49$, so it is clear that $x = 3$ can't be the vertex, as the y values for $x = 2$ and $x = 4$ aren't the same. The vertex will be at $x = -3$, as

$$y = ((-3) + 3)^2$$

$$y = (0)^2$$

To check, $x = -4$, $y = 1$, should be equal to, $x = -2$, $y = 1$. So the vertex is at $(-3, 0)$. The student most likely forgot the negative in front of the 3.

4. **Reverse engineer: a parabola has vertex $(2, -1)$ and passes through $(0, 7)$. Find its equation in the form $y = a(x - h)^2 + k$**

If the vertex is at $(2, -1)$, then $h = 2$ and $k = -1$. To complete the equation we need to find a , plugging in the values for the known point $(0, 7)$,

$$7 = a(0 - 2)^2 - 1$$

$$8 = 4a$$

$$a = 2$$

So the final form of the equation would be,

$$y = 2(x - 2)^2 - 1$$

5. **Synthesis, one sentence each: describe the full journey from $y = x^2$ to $y = -(x - 4)^2 + 2$ - every knob in order, and where the vertex ends up.**

The -4 means the vertex starts at $x = 4$. The 2 means the vertex starts at $y = 2$. The negative out the front of the parathesis means the parabola is inverted, but there isn't a number, so the graph isn't stretched or compressed. The vertex is at $(4, 2)$.